

# NEGIN FOROUZESH

Ph.D., Associate Professor

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Website: <https://neginforouzesh.github.io/>

## OCCUPATION

- Associate Professor Aug. 2025- Present  
Department of Computer Science at California State University, Los Angeles  
Principle investigator of Computational Molecular Biology (COMB) Lab.
- Assistant Professor Aug. 2020- Aug. 2025  
Department of Computer Science at California State University, Los Angeles

Expertise: Biomolecular Simulation and Modeling, Computer-Aided Drug Design, Machine Learning, Optimization, Scientific Computing, High Performance Computing, Quantum Computing.

## EDUCATION

- Ph.D. in Computer Science 2015- 2020  
Virginia Polytechnic Institute and State University (Virginia Tech)  
Blacksburg, VA, USA  
Advisor: Dr. Alexey Onufriev  
Thesis: Efficient Biomolecular Computations Towards Applications in Drug Discovery
- Master's Degree in Computer Science 2011- 2013  
Amirkabir University of Technology (Tehran Polytechnic)  
Tehran, IRAN
- Bachelor's Degree in Computer Science 2007- 2011  
Amirkabir University of Technology (Tehran Polytechnic)  
Tehran, IRAN

## RESEARCH GRANTS

- **(single PI): NIH SuRE FIRST R16**, "Improving the Accuracy of Implicit Solvents with a Physics-Guided Neural Network", 2022-2026, Total: \$730,000 [Supplement Award]: "A Workstation for Running Molecular Simulations", 2023-2024, Total: \$24,632
- **(co-PI): NSF PREC CHE Standard Grant**, "Cal State LA - MolSSI PREC Pathway to Diversity Program", 2022-2025, Total: \$886,474.00
- **(single PI): NSF CISE IIS Standard Grant**, "Workshop on Computational Structural Biology 2021", 2021-2022, Total: \$30,772
- **(single PI): CSUPERB New Investigator Grant Program**, "Computational Study of Novel Coronavirus Mutant Binding to the Human ACE2 Receptor", 2021-2022, Total: \$14,997. Acceptance Rate: %36.

## AWARDS & HONORS

- **Graduate Student of the Year Award** Mar. 2020  
Recognized by the Virginia Tech Graduate School based on academic achievements, service contributions, and commitment to advancing women in science.
- **Computer Science Scholars and Pratt Fellowships** 2017, 2019  
Exceptional scholar admitted at Computer Science Department, Virginia Tech.
- **ACM Student Research Competition** Sep. 2018  
Third place award in graduate students competition track, GHC 2018.

## PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant** Jan. 2019- June 2020  
I worked on “Accurate Yet Fast Implicit Solvation” research project funded by NIH (R21) at Virginia Tech. My role was to perform optimizations on biomolecular surfaces to obtain an accurate estimation of binding free energies.
- **HPC Technical Support** Jan. 2018- Jan. 2019  
Advanced Research Computing (ARC), Virginia Tech  
I was mainly in charge of providing HPC consultancy and support for ARC users. Moreover, I built, tested and deployed software packages on clusters.
- **Biomedical Research Intern** May 2018- Aug. 2018  
Stanford Center for Genomics and Personalized Medicine (SCGPM), Stanford University  
My role was to automate data import from large number of public annotation datasets to Google Cloud Platform (GCP) and keep the datasets up-to-date without human intervention.

## Publication

Peer-Reviewed Journal Papers [\* indicates the first student author]:

1. **Forouzes, N.**, Ghafouri, F., Tolokh, I. S., & Onufriev, A. V. (2024). Optimal Dielectric Boundary for Binding Free Energy Estimates in the Implicit Solvent. *Journal of Chemical Information and Modeling*, 64(24), 9433-9448. [IF: 5.7]
2. Rishch, A.\*, Rebel, A., Nerenberg, P. S., & **Forouzes, N.** (2024). Calculation of protein-ligand binding entropies using a rule-based molecular fingerprint. *Biophysical Journal*. [IF: 3.2]
3. Mishra, S.\*, Corro-Flores, M., Krum, D., & **Forouzes, N.** (2024). Molecular Docking Improved With Human Spatial Perception Using Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics*. [IF: 5.2]
4. Bass, L.\*, Elder, L. H., Folescu, D. E., **Forouzes, N.**, Tolokh, I. S., Karpatne, A., & Onufriev, A. V. (2023). Improving the Accuracy of Physics-Based Hydration-Free Energy Predictions by Machine Learning the Remaining Error Relative to the Experiment. *Journal of Chemical Theory and Computation*. [IF: 5.5]
5. Case, D. A.,..., **Forouzes, N.**, ... & Merz Jr, K. M. (2023). AmberTools. *Journal of Chemical Information and Modeling*. [IF: 5.6]
6. Cain\*, S., Rishch, A., & **Forouzes, N.** (2022). A Physics-Guided Neural Network for Predicting Protein-Ligand Binding Free Energy: From Host-Guest Systems to the PDBbind Database. *Biomolecules*, 12(7), 919. [IF: 5.5]
7. **Forouzes, N.**, & Mishra, N. (2021). An Effective MM/GBSA Protocol for Absolute Binding Free Energy Calculations: A Case Study on SARS-CoV-2 Spike Protein and the Human ACE2 Receptor. *Molecules*, 26(8), 2383. [IF: 4.6]
8. **Forouzes, N.**, Mukhopadhyay, A., Watson, L. T., & Onufriev, A. V. (2020). Multidimensional Global Optimization and Robustness Analysis in the Context of Protein-Ligand Binding. *Journal of Chemical Theory and Computation*. [IF: 6.006]
9. **Forouzes, N.**, Izadi, S., & Onufriev, A. (2017, August). Grid-based Surface Generalized Born Model for Calculation of Electrostatic Binding Free Energies, *Journal of Chemical Information and Modeling*. 2017, 57, 2505-2513. [IF: 3.804]

Peer-Reviewed Conference Papers [\* indicates the first student author]:

1. Thomas, L. B.\*, Rishch, A., & **Forouzes, N.** (2025, October). Multi-objective loss function for free energy calculations. *Proceedings of the 16<sup>th</sup> International Conference on Bioinformatics, Computational Biology and Health Informatics* (pp. 1-10).
2. Dhiman, N.\*, Sagar, D., & **Forouzes, N.** (2025, October). Physics-Guided Active Learning for New Ligand Discovery. In *Companion Proceedings of the 16<sup>th</sup> International Conference on Bioinformatics, Computational Biology and Health Informatics* (pp. 1-7).
3. Sagar, D.\*, Rishch, A., Sheikh, N., & **Forouzes, N.** (2023, September). Physics-Guided Deep Generative Model For New Ligand Discovery. In *Proceedings of the 14<sup>th</sup> ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics* (pp. 1-9).
4. Rebel, A.\*, Rishch, A., Massoudian, N., & **Forouzes, N.** (2022, December). Calculating the Binding Entropy of Host-Guest Systems with Physics-Guided Neural Networks. *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 3478-3485). IEEE.
5. Ramirez, S.\*, Brieu, M., & **Forouzes, N.** (2022, December). Low Data Image Analysis with a Generative Adversarial Network: A Case Study on Women Pelvic MRI Scans. *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 2690-2695). IEEE.
6. Cain, S.\*, Rishch, A., & **Forouzes, N.** (2021, December). Calculation of Protein-Ligand Binding Free Energy Using a Physics-Guided Neural Network. *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 2487-2493). IEEE.
7. Zhou, B.\*, Mohammadi, F., Lim, J. S., **Forouzes, N.**, Ghasemzadeh, H., & Amini, N. (2021). Analysis of macular thickness deviation maps for diagnosis of glaucoma. In *Advances in Visual Computing: 16<sup>th</sup> International Symposium, ISVC 2021*, Virtual Event, October 4-6, 2021, Proceedings, Part II (pp. 53-64). Springer International Publishing.
8. **Forouzes, N.** (2020, September). Binding Free Energy of the Novel Coronavirus Spike Protein and the Human ACE2 Receptor: An MMGB/SA Computational Study. In *Proceedings of the 11<sup>th</sup> ACM International Conference on Bioinformatics, Computational Biology and Health Informatics* (pp. 1-7).
9. **Forouzes, N.**, Watson, L. T., and Onufriev, A. V. (2020, May). Robustness of multidimensional optimization outcomes: a general approach and a case study. *2020 Spring Simulation Conference (SpringSim)* (pp. 1-12). IEEE.
10. **Forouzes, N.**, Kazemi, M. R., and Mohades, A. (2014, June). Structure-Based Analysis of Protein Binding Pockets Using Von Neumann Entropy. *International Symposium on Bioinformatics Research and Applications* (pp. 301-309). Springer International Publishing.

Book Chapters & Editorials [\* indicates the first student author]:

1. Sagar, D.\*, Jasko, A., & **Forouzes, N.** (2024, November). Enhancing Drug Discovery via Physics-Guided Deep Generative Models. In *Computational Structural Bioinformatics Workshop* (pp. 16-30). Cham: Springer Nature Switzerland.
2. **Forouzes, N.**, & Al Nasr, K. (2023). Special Issue "Protein Modeling and Simulation: Selected Articles from the Computational Structural Bioinformatics Workshop 2021". *Biomolecules*, 13(3), 408.

- Mishra, N.\* and **Forouzes, N.**, (2022). Protein-Ligand Binding with Applications in Molecular Docking. In Algorithms and Methods in Structural Bioinformatics, Springer, 1–16.

Presentations:

- The Biophysical Society (BPS) Annual Meeting  
San Francisco, CA. [**Travel Award**] Feb. 2026
- The Biophysical Society (BPS) Annual Meeting  
Los Angeles, CA Feb. 2025
- The ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)  
Houston, TX. [**Best Paper Finalist Award**] Sep. 2023
- The American Chemical Society (ACS) Fall National Meeting  
San Francisco, CA Aug. 2023
- The Biophysical Society (BPS) Annual Meeting  
San Diego, CA Feb. 2023
- The 12<sup>th</sup> Southern California System Biology Symposium  
(SoCalSysBio 23)  
University of Southern California, CA. [**Invited Speaker**] Apr. 2023
- The 5<sup>th</sup> Southern California Theoretical Chemistry Symposium  
(SoCal TheoChem 5)  
Riverside, CA. [**Invited Speaker**] Oct. 2022
- 36<sup>th</sup> Annual CSU Student Research Competition  
San Francisco, CA & Virtual. [**First Place Award**] May. 2022
- American Chemical Society (ACS) National Meeting  
Atlanta, GA & Virtual Aug. 2021
- The American Chemical Society (ACS) Spring National Meeting  
Virtual Apr. 2021

Professional Service & Membership:

- Panel Reviewer
  - Panelist of the NSF Mathematical and Physical Sciences (MPS) 2024
  - Panelist of the CSU Biotechnology (CSUBIOTECH) 2024, 2025
  - Ad-hoc Panelist of the Ohio Cancer Research 2023 Grant Review Summer 2023
  - Panelist of the National Science Foundation (NSF) 2022
  - Panelist of the CSU Program for Education and Research in Biotechnology (CSUPERB) 2022
  - Ad-hoc Panelist of the National Science Foundation (NSF) 2021
- Organization
  - PC Member of the 18<sup>th</sup> International Conference on Bioinformatics and Computational Biology (BICOB'26) 2026
  - PC Member of the Computational Structural Biology Workshop (CSBW) at ACM-BCB 2026 2026
  - PC Member of the Computational Structural Biology Workshop (CSBW) at ACM-BCB 2025 2025

4. PC Member of the Computational Structural Biology Workshop (CSBW), University of Massachusetts, Boston 2024
  5. PC Member of the Computational Structural Biology Workshop (CSBW) at ACM-BCB 2023 2023
  6. Session Chair of the Artificial Intelligence Techniques for BioMedicine and HealthCare (AIBH) Workshop at IEEE-BIBM 2022 2022
  7. Co-Chair of the Computational Structural Biology Workshop (CSBW) at IEEE-BIBM 2021 2021
  8. Guest Editor of the MDPI Biomolecules, Special Issue: Protein Modeling and Simulation 2021-2022
- Membership
    1. Member of the Biophysical Society (BPS) 2022-
    2. Member of the American Chemical Society (ACS) 2017-
    3. Member of the ACM Special Interest Group on Bioinformatics, Computational Biology, and Biomedical Informatics (SIGBio) 2020-2023
    4. Member of Association for Computing Machinery (ACM) 2016-2023
    5. Treasurer of “Computer Science Grad Council at Virginia Tech” 2018-2019
    6. Treasurer of the “Iranian Society at Virginia Tech (ISVT)” 2017- 2018
    7. Associate of “Virginia Tech Graduate Academy for Teaching Excellence” (VTGrATE) 2019-2020
  - Journal and Conference Reviewer
    1. Reviewer of the “Machine Learning and the Physical Sciences” Workshop at NeurIPS 2025
    2. Reviewer of the “Machine Learning and the Physical Sciences” Workshop at NeurIPS 2024
    3. Reviewer of the Journal of Chemical Theory and Computation (JCTC) 2024, 2025
    4. Reviewer of the Journal of Bioinformatics Advances 2024
    5. Reviewer of the Journal of BMC Biology 2024
    6. Reviewer of the Journal of Computational and Mathematical Biophysics 2024
    7. Reviewer of the “Machine Learning and the Physical Sciences” Workshop at NeurIPS 2023
    8. Reviewer of the PLOS ONE Journal 2023
    9. Reviewer of the Journal of Chemical Information and Modeling (JCIM) 2022
    10. Reviewer of the Journal of Molecular Biology (JMB) 2021
    11. Reviewer of the IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB) 2021, 2024
    12. Reviewer of Journal of the ChemBioChem 2021
    13. Reviewer of the Journal of Computational Biophysics and Chemistry (JCBC) 2020
    14. Reviewer of the Journal of Computers in Biology and Medicine (CIBM) 2020

## EDUCATIONAL PERFORMANCE

### • Teaching Records at Cal State LA

- CS 4635: Modeling and Simulation Spring 2026
- CS 4540: Introduction to Quantum Computing Fall 2025
- CS 4662: Advance Machine Learning and Deep Learning, Spring 2025
- CS 4635: Modeling and Simulation Spring 2025
- CS 2148: Discrete Structures Fall 2024
- CS 4635: Modeling and Simulation Spring 2024
- CS 4635: Modeling and Simulation Spring 2023
- CS 2148: Discrete Structures Fall 2022
- CS 2148: Discrete Structures Fall 2021
- CS 5112: Design and Analysis of Algorithm Spring 2021
- CS 2148: Discrete Structures Spring 2021
- CS 2148: Discrete Structures Fall 2020

### • Certificates

- Independent Improving Your Online Course (IYOC) Oct. 2021  
Quality Matters (QM)
- DELTA New Faculty Institute Workshops Nov. 2020  
American Society for Engineering Education (ASEE)
- Alt-Instruction Summer Institute Workshops July 2020  
Center for Effective Teaching and Learning (CETL) at Cal State LA

### • Student Supervision

- Current Students, Cal State LA
  1. Nikhil Dhiman, M.Sc. in Computer Science Fall 2024-
  2. Akash Saraf, M.Sc. in Computer Science Fall 2025-
  3. Lizbeth Vargas-Martinez, M.Sc. in Computer Science Fall 2025-
- Alumni, Cal State LA
  1. Sunheng Leng, B.Sc. in Computer Science Summer 2024- Fall 2024
  2. Uriel Baldesco, B.Sc. in Computer Science Summer 2024- Fall 2024
  3. Fergus Place, M.Sc. in Electrical Engineering Summer 2023- Summer 2025
  4. Libina Thomas, M.Sc. in Computer Science Spring 2024- Summer 2025
  5. Ali Risheh, M.Sc. in Computer Science Fall 2022- Spring 2024
  6. Shivam Mishra, M.Sc. in Computer Science Fall 2022- Spring 2024
  7. Dikshant Sagar, M.Sc. in Computer Science Spring 2023- Spring 2024
  8. Ari Jasko, B.Sc. in Computer Science Summer 2023- Spring 2024
  9. Isabella Perez, B.Sc. in Biology Summer 2023
  10. Missael Corro-Flores, B.Sc. in Biophysics Spring 2023
  11. Nida Sheikh, B.Sc. in Computer Science Fall 2022- Fall 2023
  12. Alles Rebel, M.Sc. in Computer Science Spring 2021- Fall 2022
  13. Deepanker Seth, B.Sc. in Computer Science Fall 2022
  14. Negin Massoudian, M.Sc. in Electrical Engineering Spring 2022- Fall 2022

15. Nikita Mishra, B.Sc. in Chemistry and Biochemistry      Spring 2022
16. Sahar Cain, M.Sc. in Computer Science      Fall 2022
- Graduate Thesis Committees, Cal State LA
  1. Mohsen Alavian, M.Sc. in Computer Science,      Spring 2026
  2. David Camacho, M.Sc. in Computer Science,      Spring 2026
  3. Eric Chan, M.Sc. in Computer Science,      Spring 2022
  4. Bingnan Zhou, M.Sc. in Computer Science,      Spring 2021
  5. Veronica Toriz, M.Sc. in Computer Science,      Spring 2021
  6. Kevin Delao, M.Sc. in Computer Science,      Fall 2020

**CONTRIBUTION  
TO THE  
UNIVERSITY**

University-Level Contributions:

- Member of the Graduate Studies Subcommittee (GSS) Fall 2022- Present  
Vice chair: Fall 2023, Chair: Spring 2024- Fall 2024
- Member of the Faculty Policy Committee (FPC)      Fall 2022- Spring 2023
- Faculty co-Director of the High Performance Computing (HPC)  
Center at Cal State LA      Spring 2022- Present
- Faculty Advisor of the Honors College      Fall 2021

College-Level Contributions:

- Member of the Research Taskforce      Fall 2021- Present
- Member of the Student Affairs Committee (SAC)      AY 2021-2022
- Member of Instructionally Related Activities (IRA)      AY 2020- 2021  
Funding Program
- Board Member of Biomedical Engineering (BME)      Fall 2020- Present  
Minor Program

Department-Level Contributions:

- Member of the Lecturer Evaluation Task Force      Fall 2021- Present  
Chair: Fall 2022- Present
- Member of the Graduate Program      Fall 2024- Present
- Member of the Faculty Search Committee      AY 2022- 2023
- Chair of the Student Affairs Committee (SAC)      Fall 2021- Spring 2024
- Member of the Industry Advisory Board (IAB)      Fall 2020- Spring 2024